# DNA as the hereditary material and primary DNA

## Lecture: DNA as the hereditary material and primary DNA

### 15 Oct 2026

### Part 1: Detailed Revision Notes

#### 1. Foundational Background & The Flow of Genetic Information

- **Central Dogma Framework:** The flow of genetic information outlines how DNA stores hereditary instructions, directing cellular function and continuity.
- **Early Historical Milestones:** Investigations into hereditary material required proving whether proteins or nucleic acids carried genetic instructions. The early isolation of DNA laid the groundwork for structural and functional discoveries.

#### 2. Experimental Evidence Identifying DNA as Genetic Material

- **Griffith's Transformation Experiments (1928):**
- Studied *Streptococcus pneumoniae* strains (smooth encapsulated virulent strains vs. rough non-encapsulated non-virulent strains).
- Demonstrated that heat-killed virulent bacteria could "transform" live non-virulent rough cells into virulent smooth cells, signaling a hereditary transforming principle.
- **In Vitro Transformation Assays:** Follow-up in vitro assays isolated cellular components to pinpoint the specific macromolecule responsible for transformation.
- **Hershey and Chase Experiments (1952):**
- Utilized bacteriophages (viruses infecting bacteria) labeled with radioactive isotopes (❑$^{32}$P for DNA, ❑$^{35}$S for protein coats).

- Proved conclusively that viral DNA—not protein—enters host bacterial cells during infection to reprogram them, confirming DNA as the universal genetic material.

### 3. Nucleic Acid Molecular Structure & Composition

- **Monomer Building Blocks:** Nucleic acids are polymers composed of nucleotide subunits consisting of a pentose sugar, a phosphate group, and a nitrogenous base.
- **Nitrogenous Bases:** Divided into purines (Adenine, Guanine) and pyrimidines (Cytosine, Thymine in DNA, Uracil in RNA).
- **Backbone Linkages:** Nucleotides polymerize via covalent **phosphodiester bonds** linking the 3′ carbon of one sugar to the 5′ carbon of the next via a phosphate group, establishing directionality (5′ to 3′).
- **Nomenclature:** Individual nucleotide units, such as deoxyadenosine monophosphate, reflect the specific sugar, base, and phosphate makeup.

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## Part 2: Assessment & Practice Questions

### Section A: Multiple-Choice Questions (MCQs)

*Choose the single best option for each question.*

**1. Which radioactive isotope was used by Alfred Hershey and Martha Chase to selectively label viral DNA in their 1952 experiments?**

- **A)** ❑$^{35}S$
- **B)** ❑$^{32}P$
- **C)** ❑$^{14}C$
- **D)** ❑$^{15}N$

**2. What was the primary conclusion drawn from Frederick Griffith's 1928 experiments with *Streptococcus pneumoniae*?**

- **A)** Proteins act as the primary hereditary material in bacteriophages.
- **B)** Heat-killed virulent bacteria release a "transforming principle" that can alter the phenotype of non-virulent strains.
- **C)** RNA is capable of self-replication without cellular machinery.
- **D)** Phosphodiester bonds are broken down by high temperatures.

**3. Which chemical linkage forms the primary covalent backbone of a single nucleic acid strand?**

- **A)** Hydrogen bond
- **B)** Peptide bond
- **C)** Phosphodiester bond
- **D)** Glycosidic bond

**4. In the structure of a deoxyribonucleotide like deoxyadenosine monophosphate, what components are directly attached to the central deoxyribose sugar?**

- **A)** A nitrogenous base and a phosphate group
- **B)** Only amino acids and carboxyl groups
- **C)** A peptide backbone and a sulfur atom
- **D)** Two independent purine rings

**5. What viral component did Hershey and Chase find remained outside the bacterial host cell following infection?**

- **A)** Viral DNA
- **B)** Viral protein coat labeled with sulfur
- **C)** Ribosomal RNA complexes
- **D)** Phosphodiester polymers

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## Section B: Multiple-Answer Questions (MAQs)

*Select all options that are correct. More than one choice may apply.*

**6. Which of the following experimental milestones or concepts were covered in the foundational study of nucleic acid genetics?**

- ☐ Griffith's in vivo transformation assays using *Streptococcus pneumoniae*
- ☐ Hershey and Chase radiolabeling experiments confirming DNA as the genetic material
- ☐ The formation and structural role of phosphodiester bonds in nucleic acid backbones
- ☐ The isolation and characterization of DNA

**7. Which characteristics accurately describe nucleic acid molecular structure?**

- ☐ Strands possess a defined 5’ to 3’ directionality.
- ☐ Nucleotides consist of a pentose sugar, a phosphate group, and a nitrogenous base.
- ☐ Phosphodiester bonds link the 3’ carbon of one sugar to the 5’ carbon of the adjacent sugar.
- ☐ Purine bases directly bond to amino acid side chains via peptide links.

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### Section C: Short Written Answer Questions

**8. Explain how the Hershey and Chase (1952) radioactive labeling experiment differentiated between DNA and proteins to prove that DNA is the genetic material.**

**9. Describe the structural significance of phosphodiester bonds in nucleic acid strands, noting how they dictate polymer orientation and backbone stability.**

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## Part 3: Solutions, Answer Keys, and Model Answers

### Answer Key: Multiple-Choice Questions (MCQs)

- **1. B (❑$^{32}$P)**
- *Explanation:* Phosphorus is found in DNA backbones but not in standard proteins, making ❑$^{32}$P ideal for labeling viral DNA. Sulfur (❑$^{35}$S) was used to label proteins.
- **2. B (Heat-killed virulent bacteria release a “transforming principle” that can alter the phenotype of non-virulent strains.)**
- *Explanation:* Griffith showed that rough non-virulent bacteria took up a transforming factor from heat-killed smooth cells, converting them into virulent strains.
- **3. C (Phosphodiester bond)**
- *Explanation:* Phosphodiester bonds connect adjacent nucleotide sugars, forming the strong covalent sugar-phosphate backbone.
- **4. A (A nitrogenous base and a phosphate group)**

- *Explanation:* A standard mononucleotide consists of a pentose sugar attached to a nitrogenous base (at the 1′ position) and one or more phosphate groups (at the 5′ position).
- **5. B (Viral protein coat labeled with sulfur)**
- *Explanation:* Hershey and Chase found that the radioactive protein ghosts remained outside the bacterial cell, whereas viral DNA entered the cytoplasm.

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## Answer Key: Multiple-Answer Questions (MAQs)

- **6. All options are correct.**
- *Explanation:* The lecture sequence thoroughly covered Griffith's transformation work, Hershey and Chase's phage experiments, early DNA isolation, and phosphodiester bond formation.
- **7. First three options are correct.** (Incorrect: Purine bases do not form peptide links with amino acid side chains; they form hydrogen bonds with complementary bases).
- *Explanation:* Nucleic acids are directional polymers built from nucleotide subunits joined by phosphodiester linkages between sugar carbons.

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## Model Answers: Short Written Answer Questions

- **Model Answer 8:** Hershey and Chase took advantage of the distinct chemical compositions of DNA and proteins. Proteins contain sulfur ($S$) but virtually no phosphorus, whereas DNA contains phosphorus ($P$) in its sugar-phosphate backbone but no sulfur. By growing bacteriophages in media containing radioactive sulfur (❑$^{35}$S) to label proteins or radioactive phosphorus (❑$^{32}$P) to label DNA, they tracked which component entered host bacteria during infection. After agitation in a blender and centrifugation, they discovered that ❑$^{32}$P (DNA) entered the bacterial cells and was passed on to progeny phages, while ❑$^{35}$S (protein) remained outside. This confirmed that DNA is the hereditary material responsible for viral reprogramming.
- **Model Answer 9:** Phosphodiester bonds are covalent linkages formed between the 5′ phosphate group of one nucleotide and the 3′ hydroxyl

group of the adjacent pentose sugar. This directional linking creates a stable, repeating sugar-phosphate backbone that protects the inner nitrogenous bases. Because synthesis and reading occur relative to these terminal ends, phosphodiester bonds establish strict structural polarity (running from a free 5’ phosphate to a free 3’ hydroxyl), which is crucial for template-directed replication, transcription, and overall nucleic acid stability.